Freelance Pricing and Invoicing Tool: Product Requirements v2

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Purpose and what changed from v1

v2 narrows the build to a pricing, quoting and invoicing tool that is correct to the cent for one owner who is not VAT registered, and moves everything else behind that core.

The goal is unchanged: a working business tool for a freelance software developer invoicing in South African rand, not a UI prototype. What changed is the order of work and nine decisions that v1 left open or got wrong.

Area	v2 decision	Why v1 changed
Build order	Pricing engine first, then estimates, quotes, invoices and payments, then reporting	v1 listed every feature as equal. A business tool must be correct before it is broad.
Authentication	One owner account, no roles, second factor required	There is one user. Roles add code and attack surface for nothing.
Live recalculation	Drafts recalculate live. Issued quotes and invoices are frozen snapshots	Editing a rate in settings must never rewrite an old invoice.
Complexity and margin	Complexity multiplier applies to fixed price mode only. Time based mode has no margin	Hour estimates already reflect complexity and the hourly rate already contains profit. Stacking either on top double counts.
VAT	Off by default and invisible. The schema carries VAT fields for later. A threshold monitor warns before registration becomes compulsory	v1 assumed a registered vendor. An unregistered supplier must not show VAT.
Quote to invoice	One accepted quote can produce several invoices (deposit, milestones, final) up to its accepted value	v1 implied one invoice per quote, which breaks deposits and milestones.

Area	v2 decision	Why v1 changed
Numbering and status	Numbers are allocated at issue with no gaps. Partially paid, paid and overdue are derived, never stored	Stored statuses drift out of sync with payments.
Daily rates	Dropped as an input. Days are shown as hours divided by a configured hours per day	Two rate inputs create two sources of truth.
Profitability	Moved to phase two. The MVP shows an effective hourly rate on every estimate	Actual cost tracking needs the core flows working first.

Operating assumptions and decisions

This is a production tool for one owner, so correctness and auditability outrank feature count.

Topic	Decision
Users	One owner account. No roles, teams or client logins. Every table still carries an owner id, so multi user access can be added later without a rewrite.
Tax status	Not VAT registered. Documents are titled "Invoice" and show no VAT lines and no VAT number. VAT mode is phase two.
Currency	ZAR only. Every document stores a currency code, but no conversion logic exists. The README states this limit plainly.
Business type	Not yet known: sole proprietor or registered company. Settings hold a legal name and an optional registration number so either works.
Dates	Africa/Johannesburg time. Due dates are calendar dates, not timestamps, so a document never becomes overdue early or late because of a time zone.
Payments	Recorded manually from bank statements. No payment gateway, bank feed or automatic reconciliation.

Topic	Decision
Not accounting software	No ledger and no tax returns. The tool reports what was invoiced and what was received, for management decisions.
Client data	Names, emails, addresses and financial documents are personal information under POPIA. The architecture and security section sets the handling rules.

Scope

The MVP is the shortest path from a client enquiry to a recorded payment. Phase two starts only after that path has carried one real quote, invoice and payment.

MVP

- Settings: business details, logo, rates, defaults, numbering, payment terms, terms and conditions, banking details
- Clients and projects
- Estimates in both pricing methods, with a live summary panel
- Quotations with draft, sent, accepted, rejected and expired states, and a stored PDF
- Invoices created from accepted quotations, several per quotation, with a stored PDF
- Payments: deposits, milestone, partial and final, with reversal
- Dashboard: quoted, confirmed, invoiced, received, outstanding, overdue, upcoming due dates, and a VAT threshold monitor
- Owner login with a second factor, backups, migrations, environment example, setup instructions and README

Phase two

- Actual costs, estimated versus actual, and project profitability
- Monthly revenue trends and reports
- Service and feature catalogue, and fixed price packages
- Recurring maintenance packages and recurring invoices
- Credit notes for corrections after payment
- VAT mode with tax invoice validation and zero rating
- Contingency folded into line prices, and branding beyond the logo
- CSV exports for your accountant, and email delivery of documents

Out of scope

- Multi currency invoicing
- Payment gateways, bank feeds and reconciliation
- Multiple users and roles
- Client portal
- Ledger, tax returns and payroll
- Time tracking timers
- Native mobile app (the web app stays fully responsive)

Money rules and pricing formulas

All money is stored as integer cents and every percentage uses decimal arithmetic, so a quotation and the invoices created from it always agree to the cent.

Money rules

- Amounts are stored as integer cents in a Postgres bigint column. No floating point number ever holds a currency amount.
- Percentages and multipliers use decimal.js with two decimal places.
- Rounding is half up to the nearest cent, applied once per line total and once per percentage step (urgency, contingency, discount). A document total is the sum of already rounded parts and is never rounded again.
- Hours are decimals with two places. Rates are stored in cents per hour.
- Every pricing function is pure: inputs in, an itemised result out, with no database, clock or UI access.

Method A: time based pricing

Step	Component	Calculation	Shown to client
1	Labour	Hours × category rate for each line, rounded to cents, then summed	Yes, as line items
2	Urgency surcharge	Urgency percent × labour, only when the urgent flag is on	Yes, as its own line
3	Contingency	Contingency percent × labour (before urgency)	Yes, as its own line

Step	Component	Calculation	Shown to client
4	Pass through expenses	Cost × (1 + markup percent) for each once off expense line	Billed amount only. Cost and markup stay internal
5	Subtotal	Steps 1 + 2 + 3 + 4	Yes
6	Discount	A percentage or a fixed amount off the subtotal, never more than the subtotal	Yes
7	Total	Subtotal minus discount	Yes

Complexity multiplies nothing in this method. Pick the project complexity to guide your hour estimate, because the hours already carry it. Recurring costs such as hosting or a maintenance package sit in their own list and are never added to the project total.

Method B: fixed price

Step	Component	Calculation
1	Deliverable price	Entered by you for each deliverable. Catalogue defaults arrive in phase two
2	Complexity adjustment	Deliverable price × the multiplier for the project complexity. All four multipliers are seeded at 1.00 until you set them, so nothing is assumed
3	Urgency, contingency, expenses, discount	The Method A rules, using the adjusted deliverable total as the base instead of labour
4	Internal check	Effective hourly rate = (total minus billed pass through expenses) ÷ your estimated hours. A warning shows when it falls below the minimum rate set in settings

You still enter estimated hours per deliverable in this method. They never appear on the quotation. They exist so the tool can tell you when a fixed price underpays you.

Worked example

This is the first automated test fixture. The rates and percentages are illustrative values chosen for the test, not suggested prices.

Line	Calculation	Amount (ZAR)
Development	40 hours × R600.00	R24,000.00
Design	10 hours × R500.00	R5,000.00
Labour	Sum of the two lines	R29,000.00
Urgency	20% × R29,000.00	R5,800.00
Contingency	10% × R29,000.00	R2,900.00
Hosting	R1,200.00 cost × 1.10	R1,320.00
Subtotal	Labour + urgency + contingency + hosting	R39,020.00
Discount	5% × R39,020.00	R1,951.00
Total	Subtotal minus discount	R37,069.00

The internal effective hourly rate is (R37,069.00 minus R1,320.00) ÷ 50 hours, which is R714.98, against a base blended rate of R580.00. That gap is what urgency and contingency add.

Rules that prevent double counting

- Contingency and urgency use labour as their base. Neither is charged on expenses, and neither is charged on the other.
- The discount is applied once, to the subtotal, and never to a single line as well.
- Markup applies to expense lines only, once.
- With VAT mode off no tax line exists. In phase two, VAT is calculated on the discounted subtotal and rounded once at document level.
- Every figure in the summary panel has an explain control that shows its formula and inputs.

Internal versus client facing

Expense cost, markup, effective hourly rate, minimum rate warnings and internal notes are internal only. The quotation builder copies a fixed whitelist of fields from the estimate, so any field added later is hidden from clients by default.

Functional requirements

Each requirement has an ID that tests and milestones refer to, and every requirement belongs to the MVP unless it says phase two.

Estimates

ID	Requirement
EST1	An estimate belongs to one project. Duplicating an estimate creates a new draft with its own lines.
EST2	The owner picks time based or fixed price. Switching method keeps the shared lines (expenses, urgency, discount) and names the lines that will be dropped before the owner confirms.
EST3	Labour lines hold a task, a category (development, design, testing, consulting, project management), hours and a rate. The rate defaults from settings and any override is visibly flagged.
EST4	Lines, including custom ones, can be added, edited, duplicated, reordered and removed.
EST5	Expense lines hold a description, internal cost, markup percent and billed amount, and are marked once off or recurring. Recurring lines are listed separately and excluded from the project total.
EST6	Fixed price lines hold a deliverable, a price and internal estimated hours.
EST7	A sticky summary panel shows live totals in two views: client facing, and internal (effective hourly rate, contingency amount, markup earned).
EST8	Every total has an explain control showing its formula and inputs.
EST9	Validation: hours and rates above zero, percentages from 0 to 100, discount no larger than the subtotal. The server runs the same Zod schemas as the form.
EST10	No estimate can be created until rates are configured. First run sends the owner to settings, so no price in the app is ever an unreviewed default.

Clients and projects

ID	Requirement
CLI1	A client holds a name, contact person, email, phone, address, optional company name and registration number, optional VAT number (stored but unused until VAT mode), and notes.
CLI2	The client profile lists projects, quotations, invoices, payments and the outstanding balance. A client that has documents is archived, never deleted.
PRJ1	A project holds a name, client, description, status (lead, quoted, active, on hold, completed, cancelled) and a deadline.
PRJ2	Scope changes are new quotations on the same project. The agreed price is the sum of its accepted quotations.
PRJ3	Phase two: actual costs, estimated versus actual, and profitability per project.

Quotations

ID	Requirement
QUO1	Created from a ready estimate. The quotation copies the whitelisted client facing fields as a snapshot. Later edits to the estimate or to settings never change it.
QUO2	States are draft, sent, accepted, rejected and expired. Allowed moves: draft to sent, and sent to accepted, rejected or expired. Only drafts are editable. Changing a sent quotation means duplicating it as a new draft.
QUO3	Expiry is derived. A sent quotation past its expiry date shows as expired without anyone updating it, and an expired quotation cannot be accepted. Duplicate it with a new expiry date instead.

ID	Requirement
QUO4	The number is assigned when the draft is sent, using the configured pattern. Drafts have no number.
QUO5	Contents: business details and logo, client details, project description and scope, deliverables and exclusions, itemised pricing, payment schedule, estimated timeline, revision policy, terms and conditions, acceptance block, issue and expiry dates.
QUO6	The payment schedule is a list of milestones whose percentages must total exactly 100. Milestone amounts use largest remainder rounding, so they sum to the quotation total to the cent.
QUO7	A PDF is generated when the quotation is sent and stored. Every later download serves that stored file.
QUO8	Accepting records the acceptance date and an optional accepted by name. Rejecting records an optional reason.

Invoices

ID	Requirement
INV1	An invoice is created from an accepted quotation by choosing a milestone, or as a custom invoice with a stated reason. Lines are copied from the quotation snapshot.
INV2	The total of all non cancelled invoices for a quotation can never exceed its accepted total. The quotation shows the invoiced and remaining amounts.
INV3	The number is assigned when the invoice is issued, with no gaps, using the configured pattern. Drafts have no number.

ID	Requirement
INV4	Contents: invoice number, invoice date, due date from the payment terms, business and client details, itemised lines, discount, total, amount paid, balance, banking details and a payment reference. The title is "Invoice". No VAT line and no VAT number appear.
INV5	Stored status is draft, issued or cancelled. Partially paid, paid and overdue are derived from payments and the due date, and shown as a badge.
INV6	An issued invoice cannot be edited or deleted. It can be cancelled only while it has no payments, and a reason is mandatory. Corrections after payment need credit notes (phase two).
INV7	A PDF is generated at issue and stored, as with quotations.

Payments

ID	Requirement
PAY1	A payment holds an invoice, amount, date received, method (EFT, card, cash, other), reference, notes and a label (deposit, milestone, final, other). The label is descriptive and never changes a calculation.
PAY2	Partial payments are allowed. A payment above the outstanding balance is rejected, and the remaining balance is shown.
PAY3	Payments are append only. A mistaken payment is reversed with a reason, which restores the balance and keeps both entries visible.
PAY4	Payments can be recorded only against issued invoices.
PAY5	A deposit is simply the first milestone invoice. There is no separate deposit mechanism.

Dashboard

Each figure has one definition, shown in the interface, so invoiced revenue is never confused with money received.

Figure	Definition
Quoted	Sum of totals of quotations that are sent and not expired
Confirmed	Sum of totals of accepted quotations
Invoiced	Sum of totals of issued invoices, excluding cancelled ones
Received	Sum of payments, excluding reversed ones
Outstanding	Invoiced minus received, on issued invoices
Overdue	Outstanding amount on invoices whose due date has passed
Upcoming	Issued invoices with a balance and a due date inside a configurable window (default 14 days)
VAT threshold monitor	Invoiced total for the trailing twelve months against the threshold set in settings, with a warning at a configurable share of it (default 80 percent)

The threshold is entered by you and must be verified with SARS. The monitor is an early warning only, because the legal test also looks forward over the next twelve months. Monthly revenue trends and estimated versus actual costs are phase two.

Settings

ID	Requirement
SET1	Business: legal and trading name, address, email, phone, optional registration number, logo.
SET2	Rates: an hourly rate per category, a minimum effective hourly rate, and hours per working day.

ID	Requirement
SET3	Defaults: urgency percent, contingency percent, quotation validity in days, payment terms in days, default milestone split, revision policy, terms and conditions, banking details.
SET4	Complexity multipliers for basic, intermediate, advanced and enterprise, seeded at 1.00.
SET5	Numbering patterns for quotations and invoices, for example QUO{YYYY}{SEQ:4}. The next number is visible and a counter can never be lowered.
SET6	VAT: a registered flag (off), the registration threshold and the warning share. The VAT number and rate fields stay hidden while the flag is off.
SET7	Changing a setting never alters an existing document. Setting changes are written to an audit log.
SET8	No rate, tax setting or business detail lives in source code. A seed file may supply examples, and the interface labels any seeded value as an editable suggestion.

Interface

- Sidebar navigation: Dashboard, Estimates, Clients, Projects, Quotations, Invoices, Reports (phase two), Settings.
- The new estimate workflow uses grouped form sections (project, labour or deliverables, expenses, adjustments) beside the sticky summary panel. On mobile the panel collapses to a bar at the bottom that expands on tap.
- Light and dark themes both ship in the MVP. Cards and panels use plain borders, with no coloured stripe on one edge.
- Empty states say what to do next, loading states use skeletons, and every failed save shows a message with a retry.
- The layout is responsive across desktop, tablet and mobile, and tables become stacked cards on narrow screens.
- Destructive actions, such as cancelling an invoice or reversing a payment, ask for a reason.

Document integrity

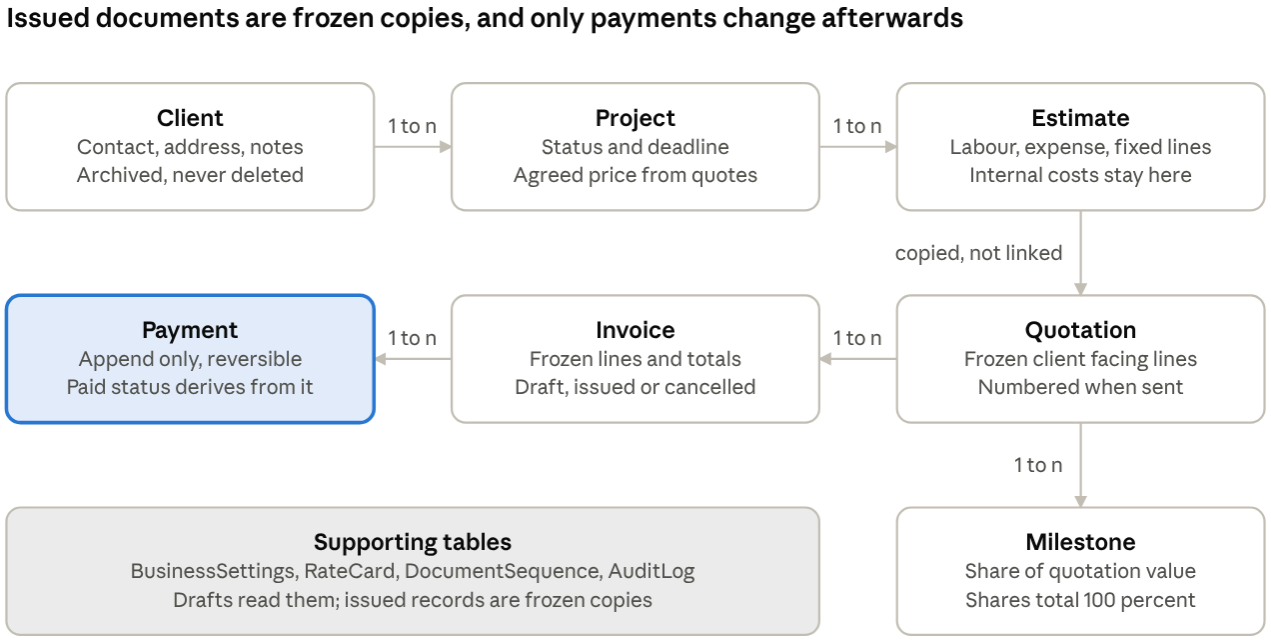
Issued documents are records, so once a quotation or invoice leaves draft it is frozen, numbered once and never recalculated.

Rule	Behaviour	How it is proven
Snapshots	At send or issue the document copies its lines, rates, totals, business details, client details, terms and banking details into its own rows. Display never reads settings again.	Change a rate after issuing. The issued totals and PDF stay identical.
Gap free numbering	The counter row is locked inside the same database transaction that issues the document. A rollback does not consume a number.	Twenty concurrent issues produce twenty unique, consecutive numbers.
Immutability	The service layer rejects edits to sent quotations and issued invoices, and a database trigger blocks updates and deletes on their financial columns.	Direct update attempts fail at both layers.
Derived status	One pure function computes paid, partially paid and overdue from payments and the due date. The interface, dashboard and tests all call it.	Boundary tests on the due date and on a zero balance.
Invoicing ceiling	A transactional check keeps the sum of invoices for a quotation at or below its accepted total.	Over invoicing is rejected, including two invoices created at the same moment.
Append only payments	Payments are never updated or deleted. A reversal is a new entry that references the original.	Reverse a payment. Both rows remain and the balance is restored.
VAT mode	Each document stores a VAT mode snapshot: none now, with standard, zero rated and exempt arriving in phase two. In none mode the VAT columns are zero, the VAT interface is hidden and the title is "Invoice".	Render tests assert that "Tax Invoice" and any VAT figure never appear in none mode output.

Rule	Behaviour	How it is proven
Audit log	Every status change, setting change and payment reversal records its time and action.	Each transition writes exactly one entry.
Stored PDFs	The PDF made at send or issue is stored and served as is, so a later template change cannot alter an old document.	Change the template, download an old invoice and compare file hashes.

Turning VAT mode on later affects new documents only. Existing documents keep the snapshot they were issued with.

Data model



entity relationships · 7 core tables, 4 supporting

Quotations and invoices copy what they need at send or issue, so editing an estimate or a setting later changes nothing that was issued. Payments are the only records that keep growing, and they are never edited. Quotations and invoices also carry the client and project ids, and every table carries an owner id.

Architecture, stack and security

The build uses Next.js with TypeScript, PostgreSQL and Prisma on Vercel, with the pricing engine as a pure TypeScript module that imports nothing from the database or the interface.

Stack

Layer	Choice	Reason
Framework	Next.js App Router, TypeScript in strict mode	It is your core stack, so effort goes into the domain and not the tooling.
Database	Managed PostgreSQL with automated daily backups	Numbering and the invoicing ceiling need transactions and row locking.
ORM and migrations	Prisma with committed migrations	Typed access and a reviewable schema history.
Validation	Zod schemas shared by forms and server actions	One definition of valid, enforced on the server.
Money	Integer cents, plus decimal.js for percentages	Removes floating point error.
Authentication	Auth.js with one owner account, a required second factor, secure httpOnly cookies and login rate limiting	The data is financial and personal.
PDF	React PDF renderer on the server. Files are stored in private object storage and served through authenticated routes	Repeatable output and stored copies.
Interface	Tailwind CSS with shadcn/ui, light and dark themes	Accessible components without custom plumbing.
Tests	Vitest for unit and property tests, Playwright for the core workflow	The testing section sets the cases.
Hosting	Vercel for the app. The database region is decided under POPIA below	Matches how you already deploy.

Layering

- Domain: pricing, money, rounding, status derivation, milestone splitting and number formatting. Pure functions that import nothing from the other layers.
- Services: transactions, authorization checks and repositories over Prisma. This is the only layer that touches the database.
- App: pages, components and server actions. A server action validates input with Zod and then calls a service. Components never contain formulas.

Security and POPIA

- Every server action and route checks the session, and every query is scoped by owner id.
- Secrets live in environment variables. The `.env.example` file lists every variable without values, and no secret is committed.
- Stored PDFs are private. Downloads go through an authenticated route or a short lived signed link.
- Logs never contain client names, emails or amounts.
- Backups run daily, and one restore is tested before launch. These records are the business, so an untested backup does not count.
- Financial records are soft deleted at most, and are kept for the period tax law requires. Verify that period with SARS or an accountant.
- Client details are personal information under POPIA and you are the responsible party. Record the hosting provider and region in the README, and confirm the rules for storing personal information outside South Africa before real client data goes in.

Testing and acceptance

A milestone is finished only when its calculations are covered by automated tests that fail when a formula changes.

Area	The tests must prove
Money	Half up rounding at the cent, no floating point drift (cases like 0.1 plus 0.2), and line totals that always sum to the document total.
Time based pricing	The worked example returns exactly R37,069.00. Contingency and urgency use labour as their base. A discount cannot exceed the subtotal.

Area	The tests must prove
Fixed price pricing	Complexity multiplies deliverable prices only. The effective hourly rate and the minimum rate warning are correct.
Property tests	Reordering lines never changes the total. Duplicating a line adds exactly that line's amount. A total is never negative.
Milestones	Percentages that do not total 100 are rejected. Amounts sum to the quotation total to the cent, including awkward totals such as one cent and thirds.
Status derivation	Issued with no payment, partially paid, paid, overdue and cancelled all resolve correctly, including a due date of today (not overdue) and a zero balance.
Transitions	Every allowed move succeeds and every other move is rejected, for quotations and invoices.
Numbering	Concurrent issuing yields unique consecutive numbers, and a rolled back issue consumes none.
Quotation to invoice	Invoices never exceed the accepted total, a milestone is invoiced once, and later rate changes leave issued totals unchanged.
Payments	Overpayment, payment on a draft and payment on a cancelled invoice are rejected. A reversal restores the balance.
Immutability	Updates and deletes on issued records fail in the service and in the database.
VAT off	Quotation and invoice output never contains tax invoice wording, a VAT number or a VAT line.
Dashboard	A seeded dataset reproduces every figure in the definitions table.
End to end	One Playwright run goes from estimate to quotation, acceptance, deposit invoice and payment, then shows the received amount on the dashboard.

MVP acceptance

- All tests pass in CI on every push, and merging is blocked when they fail.
- The domain layer has a test for every formula and every state transition.
- A real client enquiry can be taken through to a recorded payment, and the numbers match a hand calculation.
- The first three real quotations are checked by hand against the tool before it is trusted for more.
- The README explains setup, environment variables, migrations, backups, the VAT off default and the ZAR only limit.

Milestones

There are six milestones, and each ends in a gate that must pass before the next one begins. Dates are left for you to set against your weekly capacity.

#	Milestone	Delivers	Exit gate
1	Foundation and pricing engine	Repository, CI, schema and migrations, money module, pricing engine for both methods, status and milestone functions	The worked example passes and every domain test is green in CI
2	Settings and estimates	First run setup, settings, clients and projects, estimate screens with the live summary panel and explain controls	Interface totals match the domain tests, with validation on client and server
3	Quotations	Snapshot from an estimate, states, numbering at send, payment schedule, stored PDF	The numbering concurrency test passes and a sent quotation cannot be edited
4	Invoices and payments	Invoices from milestones, invoicing ceiling, derived status, payments and reversal, stored PDF	Over invoicing and overpayment are rejected, and immutability is enforced in the database
5	Dashboard and hardening	Dashboard figures, VAT threshold monitor, login with second factor, audit log, backups	The seeded dashboard test passes and a restore from backup has been tested

#	Milestone	Delivers	Exit gate
6	Launch on real work	Deployment, README, end to end test, three real quotations checked by hand	The MVP acceptance list is complete

Phase two is planned only after milestone 6, in the order your real use shows to be most painful.

Risks, open questions and items to verify

Five decisions need your answer before milestone 1, and five facts must be verified with SARS, the Information Regulator or an accountant before you rely on them.

Decisions needed from you

- [ ] Are you a sole proprietor or a registered company? It decides the legal name and registration number on documents.
- [ ] Will you invoice any client in a currency other than ZAR within the next 12 months? If yes, multi currency moves into the MVP and the plan changes materially.
- [ ] Should contingency and urgency show as separate lines on quotations (the current plan), or sit inside line prices?
- [ ] Are you comfortable hosting client data on Vercel and a managed database outside South Africa, subject to the POPIA check below?
- [ ] What numbering format do you want, and how many hours a week can you give this project, so milestone dates can be set?

To verify before relying on them

- The compulsory VAT registration threshold and the period it is measured over. The figure in your research was not checked against SARS.
- Whether services to non resident clients can be zero rated for you. This matters only when VAT mode is built.
- The required retention period for invoices and tax records.
- What a supplier that is not VAT registered may and may not print on an invoice.
- Whether you must register an information officer with the Information Regulator, and what notices clients must receive about their data.

Risks

Risk	Impact	Mitigation
Scope grows into accounting software	The MVP never ships	The out of scope list holds. Phase two starts only after a real quotation has gone through the tool.
A formula is wrong and underprices you unnoticed	Lost revenue	The worked example is a test fixture, the effective hourly rate shows on every estimate, and the first three quotations are checked by hand.
Complexity or margin is applied on top of hours	Overpricing	Complexity applies to fixed price mode only, by design.
VAT registration arrives before VAT mode is built	Compliance gap	The threshold monitor warns early and the schema already carries VAT fields.
The login is compromised	Client financial data exposed	Second factor, secure cookies, login rate limiting and no secrets in the repository.
Data is lost	Financial records gone	Daily backups, a tested restore and stored PDFs.
Regenerated PDFs differ from what was sent	Disputes with clients	PDFs are stored at send or issue and served unchanged.